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Chinese Gold Bar (CGB) Ciphers

The most notorious cipher devices exploited during the WWII era were the German Army's Enigma machines. They were at the heart of unprecedented cipher warfare, and the immediate deciphering of encrypted messages from the German army was detrimental to obtaining a superior position in battles and winning the war. Every encryption method documented in history demonstrates, to a certain extent, its own unique characteristics in the encrypted ciphertext. When the CGB cryptograms were examined visually, they gave a strong impression of being ciphertext that could have been generated by some kind of substitution cipher methods or devices, such as Enigma machines. (I didn't actually apply any mathematical or statistical analysis to them, though.) Even though the German wartime Enigma machines were not responsible for generating the CGB cryptograms, many variations of Enigma machines were manufactured for commercial applications, based on the operational principles of the German army's version in the 1930s.

The Swiss-K Enigma machine was one of those commercially available Enigma machines from that time. Switzerland was recognized as a neutral ground for international banking, allowing other nations to access banks in Switzerland without invoking conflicts with any political parties worldwide, while maintaining the secrecy of depositors from across the globe. The existence of the Swiss version of the Enigma machine suggests that records of significant and discreet banking transactions conducted by and within banks might have been encrypted to prevent accidental exposure to outsiders. It is reasonable to consider the possibility that the CGB ciphers were, in fact, generated by an Enigma machine. The Swiss-K Enigma machine would be a sensible choice for the National City Bank in China for the transaction of bank shares.

Cryptograms of CGB Ciphers

Cryptograms inscribed on the gold bars can be found at [IACR](#) and are replicated below. Among them, the cipher "HLMTAHGBGFNIV" in the original transcription was changed to "MLMTAHGBGFNIV" based on its decryption result. The first cipher letter should be not "H," but "M," and it is interpreted as "A."

How to Solve Chinese Gold Bar Ciphers

Gold Bar (with text only)	KO 08124	KO 00803 (or 00808)
JKGFIJPMCWSAEK SKCDKJCDJCYQSZKTZJPXPWIRN MQOLCSJTLGAJOKBSSBOMUPCE FEWGDRHDDEEUMFFTEEMJXZR RHZVIYQIYSXVNQXQWIOVWPJO MQOLCSJTLGAJOKBSSBOMUPCE FEWGDRHDDEEUMFFTEEMJXZR SKCDKJCDJCYQSZKTZJPXPWIRN RHZVIYQIYSXVNQXQWIOVWPJO MQOLCSJTLGAJOKBSSBOMUPCE SKCDKJCDJCYQSZKTZJPXPWIRN	(LEFT) MQOLCSJTLGAJOKBSSBOMUPCE FEWGDRHDDEEUMFFTEEMJXZR YQHUTABGALLOWLS VIOHIKNNGUAB MVERZRLQDBHQ ZUQUPNZN VIOHIKNNGUAB JKGFIJPMCWSAEK SKCDKJCDJCYQSZKTZJPXPWIRN	(MIDDLE) UGMNCBXCRLDEY RHZVIYQIYSXVNQXQWIOVWPJO SKCDKJCDJCYQSZKTZJPXPWIRN MQOLCSJTLGAJOKBSSBOMUPCE FEWGDRHDDEEUMFFTEEMJXZR (RIGHT) VIOHIKNNGUAB HFXPCQYZVATXAWIZPVE YQHUTABGALLOWLS XLYPISNANIRUSFTFWMIY KOWVRSRWTMLDH JKGFIJPMCWSAEK ABRYCTUGVZXUPB GKJFHYXODIE ZUQUPNZN GKJFHYXODIE
(UPSIDE DOWN) MLMTAHGBGFNIV ZUQUPNZN ABRYCTUGVZXUPB MVERZRLQDBHQ GKJFHYXODIE UGMNCBXCRLDEY HFXPCQYZVATXAWIZPVE	(RIGHT) MLMTAHGBGFNIV FEWGDRHDDEEUMFFTEEMJXZR RHZVIYQIYSXVNQXQWIOVWPJO HFXPCQYZVATXAWIZPVE KOWVRSRWTMLDH	

The longest cipher is "SKCDKJCDJCYQSZKTZJPXPWIRN" and it has 25 letters. The shortest cipher is "ZUQUPNZN" and it is 8 letters long.

Swiss-K Enigma Machine Simulator in Python on Jupyter Notebook

It is very difficult to get a hold of and use a real-life Swiss-K Enigma machine for solving the CGB ciphers. Instead, a virtual Swiss-K Enigma machine written in Python will be used. Jupyter Notebook is a powerful tool for running the simulator.

The codes in this project have been cloned from pyEnigma on GitHub, and they can be found here: <https://github.com/cedricbonhomme/pyEnigma>.

Information on how the Enigma machine works can be found here: https://en.wikipedia.org/wiki/Enigma_machine.

The wiring details used in Enigma machines can be found here: <https://www.cryptomuseum.com/crypto/enigma/wiring.htm>.

Basically, an Enigma machine substitutes the letter entered on the keyboard with another letter as an output result. After the substitution is completed, a light bulb associated with the output letter is lit up on the lookup panel, and the rotors are rotated into new positions, waiting for the next inputs. In this way, the machine substitutes the next input letter differently than before. Even if the same letter is entered repeatedly, the output letter changes each time. The initial starting positions for the rotors can be set up differently at the beginning and even on the fly, if desired. A rotor associates an input letter with an output letter through internal wiring. Each rotor has its own 26 internal wirings, and they are reconfigurable before use. Sometimes, more than one rotor can advance their

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positions, depending on the settings of notches. In the software simulator, several parameters have been defined to emulate the behavior of a real Enigma machine.

- Rotor Wiring
- KEY
- NOTCH
- RING
- Plug Board

The Swiss-K Enigma machine uses three rotors, one reflector, and one entry disk. Unless altered by the end user, the Swiss-K Enigma machine is shipped out with the following default settings.

Rotor	ABCDEFGHIJKLMNOPQRSTUVWXYZ	Notch
ETW	QWERTZUIOASDFGHJKPYXCVBNML	
I	LPGSZMHAEQKVXRFYBUTNICJDW	G
II	SLVGBTFXJQOHEWIRZYAMKPCNDU	M
III	CJGDPSHKTURAWZZFMYNQOVB LIE	V
UKW	IMETCGFRAYSQBZXWLHKDVUPOJN	

The simulator written in Python uses the factory settings without any customization. In other words, it is assumed that the City Bank did not alter any default settings, including plug board settings, when the CGB ciphers were generated. With those factory settings intact, there are numerous ways to set the runtime configuration of KEY, RING, and NOTCH to hide information in ciphers. Without knowing the exact settings of KEY, RING, and NOTCH, it is extremely difficult to decode the ciphers and restore the original texts. It is true that altering the factory configuration can provide a higher degree of complexity to the encrypted messages, but it would also be a burden to keep the record of configuration changes and runtime settings straight for later decryption because the counterpart, i.e., the depositor in the case of CGB ciphers, also should have the same information on the ciphers they are about to decipher.

Furthermore, it is assumed that the NOTCHes were unchanged as well. This leaves us with the understanding that KEY and RING settings are the only changeable parameters involved in the CGB ciphers. Each cryptogram in the CGB ciphers would have had a different combination of KEY and RING settings when generated.

The First Crack: A Pure Luck

The longest CGB cipher consists of 25 alphabetic letters. The shortest cipher is 8 letters long. There are all 26 alphabet letters on the rotor of the Swiss-K Enigma machine. If the starting KEY value on the first rotor is "G," the only moving rotor will be the first one during the encryption, and the remaining rotors won't move at all. When the encryption crosses "F" of the KEY of the first rotor, the first and second rotors will move one

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notch. As the second rotor has to move, it will happen only once at most since none of the CGB ciphers exceeds 26 letters. Since the third rotor never moves for the CGB ciphers, both the third rotor and the reflector can be combined into one new reflector for its wiring. In case the second rotor doesn't move, both the second and third rotors and the reflector altogether can be replaced with a single reflector.

There is no objective reason why I decided to start with the FEW23 cipher: "FEWGDRHDDEEUMFFTEEMJXZR." With the RING set to "AAA," the FEW23 cipher was decrypted by varying the KEY from "AAA" to "ZZZ," generating 17,576 texts. I ran a string match operation on each output to see if it contained the word "BANK." The following is an excerpt of the output.

```
In [5]: findWordByFixedRing(["FEWGDRHDDEEUMFFTEEMJXZR"], ["BANK"], "AAA")
Key-Ring-Cipher tuple(s) found: 4

Out[5]: [{ 'Key': 'CVU',
  'Ring': 'AAA',
  'Message': 'MKXBFBANKODXXRTMOQTDHQD',
  'Cipher': 'FEWGDRHDDEEUMFFTEEMJXZR' },
  { 'Key': 'FHG',
  'Ring': 'AAA',
  'Message': 'DWQRBZTZOADREBANKQTZQMD',
  'Cipher': 'FEWGDRHDDEEUMFFTEEMJXZR' },
  { 'Key': 'JQE',
  'Ring': 'AAA',
  'Message': 'OUSTGOVPBANKIZMUBMHUVOG',
  'Cipher': 'FEWGDRHDDEEUMFFTEEMJXZR' },
  { 'Key': 'KXB',
  'Ring': 'AAA',
  'Message': 'BANKUPCWXJKZEZPFVJRLDI',
  'Cipher': 'FEWGDRHDDEEUMFFTEEMJXZR' }]
```

OUST GOV P BANK IZMUBMHU VOG
The reverse of GOV

Four messages in total contain the word "BANK." When the KEY reached "JQE" with the RING set to "AAA," the Swiss-K simulator decoded the FEW23 cipher into "OUSTGOVPBANKIZMUBMHUVOG," which can be rendered in a more readable format as "OUST GOV P BANK IZMUBMHU VOG" with spaces inserted. It was pure luck to obtain both "OUST GOV" and "VOG" simultaneously in addition to "BANK." The movement of the rotors is sequential, meaning that having "VOG" decoded at the end of the FEW23 cipher completes the decipherment result and "IZMUBMHU," or IZU08 cipher in short, is a valid cipher embedded within another cipher. In other words, IZU08 is not random but intended.

This is the first evidence that the CGB cipher is a multiply enciphered type. This marks the first breakthrough in the decipherment of CGB ciphers. The "P" could stand for Purchase or something else like Private. "OUST GOV" indicates the buyer as an anonymous party from the ousted government who purchased shares in the National City Bank. A decrypted text of the FEW23 cipher has finally been produced on the Swiss-K Enigma machine simulator, with a KEY-RING pair set to "JQE" and "AAA."

Swiss-K Enigma Machine: The One

The Swiss-K Enigma machine uses three rotors, and each rotor has 26 internal wirings to pass the incoming electric signals through to the next wheel. When a letter is pressed down, the electric current passes through ROTOR-I, ROTOR-II, and ROTOR-III. The output signal of ROTOR-III is bounced back by the non-rotating fixed reflector and is sent back up to ROTOR-I.

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Finally, the output current of ROTOR-I activates the light bulb connected to the corresponding letter. For each letter entered, two forward and backward wires of the rotor are involved in the signal activation. As every letter of the FEW23 cipher is deciphered from start to end, 24 wires of ROTOR-I, 24 wires of ROTOR-II, 24 wires of ROTOR-III, and 12 wires of the reflector are involved in conducting electric signals. That is 84 out of 91 wires in total, which accounts for 92.3% usage in the decipherment of the FEW23 cipher.

```

"FDGASBDEHHEPTEHJZLH", "out": "00079598AKHJYHSHHWHV", key: "FDG", "name": "ROTOR_C", "id": "ROTOR_C_Roleplaying", "LP": "G2H4H6G9V3H5H7H9H10H11H12H13H14H15H16H17H18H19H20H21H22H23H24H25H26H27H28H29H30H31H32H33H34H35H36H37H38H39H40H41H42H43H44H45H46H47H48H49H50H51H52H53H54H55H56H57H58H59H60H61H62H63H64H65H66H67H68H69H70H71H72H73H74H75H76H77H78H79H80H81H82H83H84H85H86H87H88H89H90H91H92H93H94H95H96H97H98H99H100H101H102H103H104H105H106H107H108H109H110H111H112H113H114H115H116H117H118H119H120H121H122H123H124H125H126H127H128H129H130H131H132H133H134H135H136H137H138H139H140H141H142H143H144H145H146H147H148H149H150H151H152H153H154H155H156H157H158H159H160H161H162H163H164H165H166H167H168H169H170H171H172H173H174H175H176H177H178H179H180H181H182H183H184H185H186H187H188H189H190H191H192H193H194H195H196H197H198H199H200H201H202H203H204H205H206H207H208H209H210H211H212H213H214H215H216H217H218H219H220H221H222H223H224H225H226H227H228H229H230H231H232H233H234H235H236H237H238H239H240H241H242H243H244H245H246H247H248H249H250H251H252H253H254H255H256H257H258H259H260H261H262H263H264H265H266H267H268H269H270H271H272H273H274H275H276H277H278H279H280H281H282H283H284H285H286H287H288H289H290H291H292H293H294H295H296H297H298H299H300H301H302H303H304H305H306H307H308H309H310H311H312H313H314H315H316H317H318H319H320H321H322H323H324H325H326H327H328H329H330H331H332H333H334H335H336H337H338H339H340H341H342H343H344H345H346H347H348H349H350H351H352H353H354H355H356H357H358H359H360H361H362H363H364H365H366H367H368H369H370H371H372H373H374H375H376H377H378H379H380H381H382H383H384H385H386H387H388H389H390H391H392H393H394H395H396H397H398H399H400H401H402H403H404H405H406H407H408H409H410H411H412H413H414H415H416H417H418H419H420H421H422H423H424H425H426H427H428H429H430H431H432H433H434H435H436H437H438H439H440H441H442H443H444H445H446H447H448H449H450H451H452H453H454H455H456H457H458H459H460H461H462H463H464H465H466H467H468H469H470H471H472H473H474H475H476H477H478H479H480H481H482H483H484H485H486H487H488H489H490H491H492H493H494H495H496H497H498H499H500H501H502H503H504H505H506H507H508H509H510H511H512H513H514H515H516H517H518H519H520H521H522H523H524H525H526H527H528H529H530H531H532H533H534H535H536H537H538H539H540H541H542H543H544H545H546H547H548H549H550H551H552H553H554H555H556H557H558H559H560H561H562H563H564H565H566H567H568H569H570H571H572H573H574H575H576H577H578H579H580H581H582H583H584H585H586H587H588H589H590H591H592H593H594H595H596H597H598H599H600H601H602H603H604H605H606H607H608H609H610H611H612H613H614H615H616H617H618H619H620H621H622H623H624H625H626H627H628H629H630H631H632H633H634H635H636H637H638H639H640H641H642H643H644H645H646H647H648H649H650H651H652H653H654H655H656H657H658H659H660H661H662H663H664H665H666H667H668H669H670H671H672H673H674H675H676H677H678H679H680H681H682H683H684H685H686H687H688H689H690H691H692H693H694H695H696H697H698H699H700H701H702H703H704H705H706H707H708H709H710H711H712H713H714H715H716H717H718H719H720H721H722H723H724H725H726H727H728H729H730H731H732H733H734H735H736H737H738H739H740H741H742H743H744H745H746H747H748H749H750H751H752H753H754H755H756H757H758H759H760H761H762H763H764H765H766H767H768H769H770H771H772H773H774H775H776H777H778H779H780H781H782H783H784H785H786H787H788H789H790H791H792H793H794H795H796H797H798H799H800H801H802H803H804H805H806H807H808H809H810H811H812H813H814H815H816H817H818H819H820H821H822H823H824H825H826H827H828H829H830H831H832H833H834H835H836H837H838H839H840H841H842H843H844H845H846H847H848H849H850H851H852H853H854H855H856H857H858H859H860H861H862H863H864H865H866H867H868H869H870H871H872H873H874H875H876H877H878H879H880H881H882H883H884H885H886H887H888H889H890H891H892H893H894H895H896H897H898H899H900H901H902H903H904H905H906H907H908H909H910H911H912H913H914H915H916H917H918H919H920H921H922H923H924H925H926H927H928H929H930H931H932H933H934H935H936H937H938H939H940H941H942H943H944H945H946H947H948H949H950H951H952H953H954H955H956H957H958H959H960H961H962H963H964H965H966H967H968H969H970H971H972H973H974H975H976H977H978H979H980H981H982H983H984H985H986H987H988H989H990H991H992H993H994H995H996H997H998H999H1000H1001H1002H1003H1004H1005H1006H1007H1008H1009H1010H1011H1012H1013H1014H1015H1016H1017H1018H1019H1020H1021H10
```

The dark colors in the tables indicate the wires of each rotor that are involved in passing the signals. Each row of a table shows the wiring depending on the KEY position. Since the rotor advances one position per key press, the rotor wiring shifts left by one.

Three tables on the left indicate the forward signal transmission. The table at the bottom is for the reflector. Three tables on the right show the wirings used for the backward signal mappings. Based on the decoded text of the FEW23 cipher, it would be safe to say that the Swiss-K Enigma machine was the one involved in the generation of the CGB ciphers. We can now focus on the search for KEY-RING pairs and leave everything else out of the way.

Self Sufficient Cipher: Last Three Letters (L3L)

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When deciphering the CGB cipher, two parameter values need to be known to both parties, the sender and receiver: the initial KEY and the RING values. The KEY value changes after every keystroke, but the RING values don't change once configured. In the FEW23 decipherment, the RING was fixed at the value of "AAA." It would be interesting to see whether the same RING setting applies equally to other CGB ciphers. However, it seems very unlikely.

It is necessary for the sending party of ciphers to memorize the KEY and RING values they used for each cipher, and the receiving party needs to know those values in advance to decode the ciphers. At this point, I bluntly assumed: what if each CGB cipher conveys the RING value within it? I devised a routine to extract RING values from the given cipher and used them to see if they produced the same results. For the FEW23 cipher, all possible combinations of three consecutive letters were extracted from the string in both directions: forward and backward. Since we already know that the FEW23 cipher contains "OUSTGOV," it is used as a target string for the search. For each three-letter RING value, the FEW23 cipher was decoded by varying KEY values from "AAA" to "ZZZ." The following are the results.

```
--- Forward ---
['FEW', 'EWG', 'WGD', 'GDR', 'DRH', 'RHD', 'HDD', 'DDE', 'DEE', 'EEU', 'EUM', 'UMF', 'MFF', 'FFT', 'FTE', 'TEE', 'EEM', 'EMJ', 'MJX', 'JXZ', 'XZR']

FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUVOG -key- GPV -ring- XZR
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- MHL -ring- DRH
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- MTL -ring- DDE
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- MUI -ring- DEE
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- NMK -ring- EWG
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- NUY -ring- EEU
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- NKQ -ring- EUM
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- NUQ -ring- EEM
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- NCN -ring- EMJ
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMDKWE -key- OUA -ring- FEW
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMDKWE -key- OVX -ring- FFT
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMDKWE -key- OJI -ring- FTE
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBSDKWKE -key- PTV -ring- GDR
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBSDKWKE -key- QTH -ring- HDD
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZVHSDKWKE -key- SND -ring- JXZ
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKELMVDHSDKWKE -key- VVJ -ring- MFF
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKELMVDHSDKWKE -key- VZB -ring- MJX

--- Reverse ---
['RXZ', 'ZXJ', 'XJM', 'JME', 'MEE', 'EET', 'ETF', 'TFF', 'FFM', 'FMU', 'MUE', 'UEE', 'EED', 'EDD', 'DDH', 'DHR', 'HRD', 'RDG', 'DGW', 'GWE', 'WEF']

FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUVOG -key- GZQ -ring- XJM
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUVOG -key- INN -ring- ZXJ
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- MTL -ring- DDH
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- MXV -ring- DHR
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- MWA -ring- DGW
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- NUX -ring- EET
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- NJJ -ring- ETF
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- NUH -ring- EED
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMHUWKE -key- NTH -ring- EDD
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMDKWE -key- OVQ -ring- FFM
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBMDKWE -key- OCY -ring- FMU
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBSDKWKE -key- PMI -ring- GWE
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZMUBSDKWKE -key- QHH -ring- HRD
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKIZVHSDKWKE -key- SCI -ring- JME
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKELMVDHSDKWKE -key- VUI -ring- MEE
FEWGRHDDEEUMFFTEEMJXZR: OUSTGOVPBANKELMVDHSDKWKE -key- VKI -ring- MUE
```

Seventeen cases out of all three consecutive letters extracted from the FEW23 cipher produced the same deciphered texts. After reversing the FEW23 string, three consecutive letters were extracted, and sixteen cases also produced the same text "OUSTGOV". As you can see, there are all three pairs of KEY and RING that produce the exact same decrypted text. It is surprising to see that there are so many KEY-RING pairs that yield the same result. The most interesting one is the RING value of "XZR," which is the L3L of the FEW23 cipher.

The idea of using the last three letters (L3L) of the cipher as the RING is absolutely brilliant, if it works every time, due to the realization of how one could possibly know what the L3L of the encrypted text would be before

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having encrypted it. It is uncertain how it is achievable to determine the L3L of the encrypted text from its plaintext in advance. I don't have an answer to this requirement, but I think it might be feasible if a RING value were assigned arbitrarily in advance and one then tried to find out if there is any KEY value that could produce the L3L of the cipher coincidentally identical to the RING used in the encryption. This led me to the next simulation.

In the simulation, the plaintext to encrypt is "HELLOWORLD." I arbitrarily picked two different RING values to obtain the L3L at the end of the ciphertext. They are "XZR" and "XXX," respectively.

```
In [5]: ring = "XZR"
msg = "HELLOWORLD"
msgToCipherByRing(msg, ring, show=True)

There is no Key-Ring-Cipher tuple with the designated ring, XZR

Out[5]: []

In [6]: ring = "XXX"
msg = "HELLOWORLD"
msgToCipherByRing(msg, ring, show=True)

Key-Ring-Cipher tuple(s) found: 2
{'Key': 'CFY', 'Ring': 'XXX', 'Message': 'HELLOWORLD', 'Cipher': 'RBTUTBUXXX'}
{'Key': 'HDO', 'Ring': 'XXX', 'Message': 'HELLOWORLD', 'Cipher': 'WXMARVWXXX'}

Out[6]: [{'Key': 'CFY',
          'Ring': 'XXX',
          'Message': 'HELLOWORLD',
          'Cipher': 'RBTUTBUXXX'},
         {'Key': 'HDO',
          'Ring': 'XXX',
          'Message': 'HELLOWORLD',
          'Cipher': 'WXMARVWXXX'}]
```

The RING of "XZR" didn't produce any cipher ending with "XZR." The RING of "XXX" produced two encrypted texts ending with "XXX," and the KEY values were obviously different. Either one can be chosen for the encryption of "HELLOWORLD." Now, the cipher carries the RING value with which the original text can be produced in the decryption. The simulation result indicates that knowing the RING value doesn't specify the KEY for that cipher.

Since the KEY value can't be assigned arbitrarily, encapsulating the KEY value in the encrypted message would be impossible. This implies that both the sender and receiver in the exchange of secret texts need to know which KEY value was used in the given ciphers. It also implies that there could be predetermined KEY-RING pairs shared by each party, and each party uses their own KEY-RING pair to exchange information, including the KEY value beforehand. By the way, it is surprising how they calculated the L3L without using the computing power of modern computer systems.

Decoding IZU08 Cipher: The Completion of FEW23 Cipher

The IZU08 cipher ("IZMUBMHU") is part of the decoded FEW23 cipher. It appears random, but it could be another encrypted text itself. So, let's find out. It is assumed that the L3L "MHU" was used as the RING value for the encryption of the IZU08 cipher. Since it is unknown which KEY value was used to create the cipher, we have to collect deciphered texts with all the KEY values from "AAA" to "ZZZ." This produces a list of 17,576 texts, and

each decrypted text needs to be examined to narrow it down to a manageable size and finalize the plausible text. It is noted that this process is inevitable and prone to human error.

```
|00003 - CT GO LEES : CTGOLEES: KEY - WSF: RING - MHU -> CT (Bank) GO LEES
```

```

00002 - AYERS JOP : AYERSJOP: KEY - ACK: RING - MHU
                                KEY Value   RING Value
                                Decrypted Message
                                Dec. Message with
                                spaces inserted
                                # of words

```

In summary, the following points are verified:

- The Swiss-K Enigma machine with the factory configuration was the device used in generating CGB ciphers.
- The original NOTCH values were unchanged.
- The original ROTOR and reflector wirings were not altered.
- The L3L of a CGB cipher is used as the RING value for its decipherment unless instructed otherwise.
- The CGB cipher had plain text encrypted, and another encrypted text can be included as part of the original text. In other words, the CGB ciphers allow multiple encryptions.

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- It would be highly unlikely for two different CGB ciphers to use the same set of KEY and RING values, as it is not guaranteed that the same RING value will be available to both ciphers simultaneously.

Deciphering the CGB ciphers is straightforward from this point on.

HTG12 (HUD TAB GALLOW) Cipher: A Place Holder for another Cipher

On the left side of the Gold Bar labeled with KO 08124, the following transcription is inscribed.

```
MQO24: MQOLCSJTLGAJOKBSSBOMUPCE
FEW23: FEWGDRHDDEEUMFFTEEMJXZR
YQH16: YQHUDTABGALLOWLS
VIO12: VIOHIKNNGUAB
MVE12: MVERZRLQDBHQ
ZUQ08: ZUQUPNZN
VIO12: VIOHIKNNGUAB
JKG14: JKGFIJPMCWSAEK
SKC25: SKCDKJCDJCYQSZKTZJPXPWIRN
```

The YQH16 cipher contains plain text in the middle, which is "HUDTABGALLOW" or "HUD TAB GALLOW" (HTG12). It starts with "YQ" and ends with "LS." It is highly unlikely to obtain the YQH16 cipher from a plain text. If both "YQ" and "LS" were plug board settings, the decryption of the FEW23 cipher would change to "OUL**T** GOV P BANK CT GO**S** EEL VOG." This breaks plausibility. They couldn't be any plug board settings. (If they were intended for plug board settings, the cipher text would have been different to start with. So, it is not just a plain replacement shown here, but I think it conveys the idea.) Its length of the HTG cipher is 12 letters wide, which is the same as the length of the VIO12 cipher. The YQH16 cipher is not a cipher itself; it is regarded as a placeholder. The puzzle can be solved when the MVE12 cipher is decrypted first. The MVE12 cipher is decrypted as follows.

```
00005 - TWO MSG XOR S RH : TWOMSGXORSRH: KEY - GIV: RING - BHQ
```

With the KEY of "GIV" and the RING of "BHQ," the MVE12 cipher is translated into "TWO MSG XOR SRH." The two messages referred to by the MVE12 cipher are the YQH16 and VIO12 ciphers. The operation of XOR with those two messages produces a new YQV16 cipher.

```
YQV16: YQVIOHIKNNGUABLS
```

It is assumed that "SRH" at the end is used as the RING value for the YQV16 cipher. (Actually, I tried to obtain the deciphered text of the YQV16 cipher with the RING of "BLS," but it didn't produce any plausible results. It is also possible that I missed other plausible text, though.) The decipherment of the YQV16 cipher with "SRH" as the RING value produced the following result.

```
00005 - CONTACT FW T DEW VRY : CONTACTFWTDEWVRY: KEY - ZDE: RING -
      SRH -> CONTACT [FWT] DEWVRY
```

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The deciphered text is "CONTACT FWT DEW VRY." The name "DEW" can be found on the inscriptions of gold bars, and it looks like "Dewly" or something similar. "FWT" could be a RING value for another cipher. When the length of the CGB cipher was intended to be shorter than 26 letters, the space requirement would have forced the original text to be compacted. It is expected that abbreviations will be used. "DEWVRY" could be the name of the person who was the contact point.

Since the VIO12 cipher doesn't decipher itself, the repeated occurrence of the VIO12 cipher below the ZUQ08 cipher is considered a contextual marker to indicate the separation of MVE12 from others. The YQH16 cipher is a placeholder, and it has not been decrypted as well.

Deciphering MQ024 Cipher

The decipherment of the MQ024 cipher is somewhat more complicated because it contains double-coded ciphers within it. In the first decryption, the following message was decoded and selected. (When anyone disagrees with me on this, they can perform their own decryption, different from the one presented here.)

```
00011 - USE AL TY K D BJ Z GOV DVD ULI DUD :  
        USEALTYKDBJZGOVDVDULIDUD: KEY - FWQ: RING - PCE -> USE  
        ALTYKDBJZ GOV DVDULIDUD
```

With the KEY of "FWQ" and the RING of "PCE," we have a message divided into two parts. One part starts with "USE" and the other with "GOV." The "USE" clause has a cipher labeled ALT09 ("ALTYKDBJZ"), and the "GOV" clause has a cipher labeled DVD09 ("DVDULIDUD"). Both ciphers are 9 letters long. They are decrypted as follows.

```
00003 - D JUD BORIS : DJUDBORIS: KEY - YCI: RING - BJZ  
ALT09: ALTYKDBJZ ◊ D JUD BORIS
```

```
00003 - IDP KEY SEA : IDPKEYSEA: KEY - CPU: RING - DUD  
DVD09: DVDULIDUD ◊ ID P KEY SEA
```

Altogether, the MQ024 cipher can be deciphered.

```
MQU24: MQOLCSJTLGAJOKBSSBOMUPCE ◊ USED JUD BORIS GOV ID P KEY SEA
```

The "P" could indicate "Purchase," as in the "P BANK" of the FEW23 cipher. Here we have another name: BORIS the JUDGE. "SEA" is the P-KEY. Otherwise, it also can be interpreted as "(who) USED [JUD BOR] IS GOV (with) ID P KEY SEA." ID stands for identification and P for purchase. In this case, the decrypted text is as follows.

```
MQU24: MQOLCSJTLGAJOKBSSBOMUPCE ◊ USED [JUD BOR] IS GOV ID P KEY SEA
```

"JUD" and "BOR" could be a KEY-RING pair for transaction-related cryptograms. There may be more cryptograms than were found on the Chinese gold bars. Nonetheless, based on the deciphered text, it might be used to identify the OUST GOV or someone on that side.

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MQO24: MQOLCSJTLGAJOKBSSBOMUPCE ◊ USED [JUD BOR] IS GOV ID P KEY SEA
FEW23: FEWGDRHDDEEUMFFTEEMJXZR ◊ OUST GOV P BANK CT GOL EES VOG
YQH16: YQ**HUDTABGALLOW**LS (place holder not decrypted)
VIO12: VIOHIKNNGUAB (not decrypted alone)
MVE12: MVERZRLQDBHQ ◊ TWO MSG XOR SRH
YQV16(YQH16 XOR VIO12) : YQ**VIOHIKNNGUAB**LS ◊ CONTACT FWT DEWVRY
ZUQ08: ZUQUPNZN
VIO12: VIOHIKNNGUAB
JKG14: JKGFIJPMCWSAEK
SKC25: SKCDKJCDJCYQSZKTZJPXPWIRN

Deciphering ZUQ08 Cipher

When ZUQ08 was deciphered with the RING of "NZN," several decoded texts could be selected based on their plausibility. For example, I narrowed them down to three texts as candidates for the decoded message.

00003 - GOD BY ILL : GODBYILL: KEY - GVC: RING - NZN
00003 - ALL OR RUE : ALLORRUE: KEY - JFR: RING - NZN
00003 - XOR ID IWA : XORIDIWA: KEY - PTB: RING - NZN

Any one of them could provide certain context to the deciphered texts and can be considered the deciphered text. The problem is that as long as the correct KEY value is unknown, none of them would be the original text.

If the ZUQ08 cipher was deciphered with the RING of "FWT" from the deciphered YQV16 cipher, the following texts were selected.

00003 - XOR ID IWA : XORIDIWA: KEY - HQH: RING - FWT
00003 - GOD BY ILL : GODBYILL: KEY - YSI: RING - FWT
00003 - ALL OR MEP : ALLORMEP: KEY - BCX: RING - FWT

It seems that "ALL OR RUE" was replaced with "ALL OR MEP." The phrase "GOD BY ILL" doesn't seem to align well with the other ciphers. The context of "GOD BY ILL" would contradict other deciphered texts. It leaves "XOR ID IWA" as the final option to choose from.

Deciphering JKG14 Cipher

With the RING of "AEK", the JKG14 cipher is decoded.

00006 - ACT DMZ EF G SIG CT : ACTDMZEFGSIGCT: KEY - CAS: RING -
AEK ◊ ACT DMZ [EFG] SIG CT

"EFG" is regarded as the RING for the next cipher SKC25. With the deciphered YQV16 cipher, the deciphered texts together indicate that the contact person DEW (or DEWVRY) acts like DMZ and signals CT Bank ("CT" from the IZU08 cipher contained in the FEW23 cipher).

Deciphering SKC25 Cipher

With the RING of "EFG" from the deciphered text of the JKG14 cipher, the SKC25 cipher can be decrypted as follows.

```
00012 - BAY W X DY FX J F EX CBN S TACKY LOT :
        BAYWXDYFXJFEXCBNSTACKYLOT: KEY - HNN: RING - EFG ◇ BAY
        WXDYFXJFEXCBN STACKY LOT
WXD13: WXDYFXJFEXCBN
```

With the RING of "CBN," the WXD13 cipher is decrypted.

```
00004 - ADJ GATA MAST WY : ADJGATAMASTWY: KEY - RPY: RING - CBN ◇ ADJ
        GATA MAST WY
```

It appears to be a street name. The final text will look like the following.

```
00012 - BAY W X DY FX J F EX CBN S TACKY LOT :
        BAYWXDYFXJFEXCBNSTACKYLOT: KEY - HNN: RING - EFG ◇ BAY [ADJ
        GATA MAST WY] STACKY LOT
```

"ADJ" could stand for "adjacent." Once again, deciphering a CGB cipher is a task of narrowing down one text from all possible 17,576 outcomes, and it doesn't guarantee that the finalized one will always be the original message. The result presented in this report is merely an example.

Deciphered Texts on the Left Column of Gold Bar (KO 08124)

All ciphers on the left column of the gold bar labeled KO 08124 are listed below with their deciphered texts.

Label	Cipher Message	KEY:RING	Deciphered Message
MQO24	MQOLCSJTLGAJOKBSSBOMUPCE	FWQ:PCE	USE <u>ALTYKDBJZ</u> GOV DVDULIDUD ◇ USED [JUD BOR] IS GOV ID P KEY SEA
◇ALT09	ALTYKDBJZ	YCI:BJZ	D JUD BORIS
◇DVD09	DVDULIDUD	CPU:DUD	ID P KEY SEA
FEW23	FEWGDRHDDEEUMFFTEEMJXZR	GPV:XZR	OUST GOV P BANK <u>IZMUBMHU</u> VOG ◇ OUST GOV P BANK CT GOL EES VOG
◇IZM08	IZMUBMHU	WSF:MHU	CT GOL EES
YQH16	YQ <u>HU</u> <u>TABGALLOW</u> LS		(A place holder)
VIO12	<u>VIOHIKNGUAB</u>		(Not deciphered as it is)
MVE12	MVERZRLQDBHQ	GIV:BHQ	TWO MSG XOR [SRH]
YQV16	YQ <u>VIOHIKNGUAB</u> LS	ZDE:SRH	CONTACT [FWT] DEWVRY
ZUQ08	ZUQUPNZN	HQH:FWT	XOR ID IWA
VIO12	VIOHIKNGUAB		(Not deciphered as it is)
JKG14	JKGFIJPMCWSAEK	CAS:AEK	ACT DMZ [EFG] SIG CT
SKC25	SKCDKJCDJCYQSZKTZJPXPWIRN	HNN:EFG	BAY <u>WXDYFXJFEXCBN</u> STACKY LOT ◇ BAY ADJ GATA MAST WY STACKY LOT
◇WXD13	WXDYFXJFEXCBN	RPY:CBN	ADJ GATA MAST WY

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Three letters enclosed in brackets are used as the RING value for the decipherment of the next cipher in line. "SRH" from the MVE12 cipher contributed to the decipherment of the YQV16 cipher. Note that "TWO MSG" indicates both YQH16 and VIO12 ciphers. By performing the "XOR" operation on them, the VIO12 cipher takes the place of "HUDTABGALLOW" in the YQH16 cipher, allowing the YQV16 cipher to be formed.

The three letters "IWA" from ZUQ08 do not apply as the RING value for the JKG14 cipher. In fact, when "IWA" was applied to the decipherment of the JKG14 cipher, it did not yield any plausible results. The presence of the VIO12 cipher just below the ZUQ08 cipher might imply that the application of "IWA" as the RING value does not extend beyond that point. This interpretation may also suggest that if there is no VIO12 cipher following the ZUQ08 cipher, the RING for the cipher next to the ZUQ08 cipher would be "IWA."

Synchronizing the RING value to the L3Ls of an encrypted cipher does not work with any randomly chosen three letters. It is easier to determine within seconds if the cipher will conclude with the RING value, aided by modern computer systems. Finding the RING value for any cipher each time would not have been an easy task in the 1930s. Occasionally, they could not determine the RING value for the encryption of text in time. Embedding the RING value within a cipher for the next cipher might serve as an alternative solution to this kind of problem. It does not require the cipher to conclude with the embedded RING value from another cipher.

Limitations on the Decryption of CGB Ciphers

If both KEY and RING values for a given CGB cipher are known, the original message can always be determined decisively. Typically, the last three letters of each CGB cipher are used as the RING value for decryption unless stated otherwise. Unfortunately, the KEY values for each CGB cipher are unknown, and the lack of information makes the recovery process very challenging since a single message must be selected from a multitude of texts as the presumed original text.

In addition to that, a CGB cipher might not consist of one plain text. It could contain other encrypted texts within the deciphered text. When the decrypted text contains only partial plain text, there may be other embedded ciphers to verify. Nonetheless, the generation process of the CGB ciphers has been reverse-engineered and detailed well in this report.

The Bottom Part of Gold Bar with Dual Text (Upside Down)

The gold bar with Chinese texts and cryptograms side by side doesn't have a label on it. The bottom part of the text consists of five lines of ciphers.

Line 1: (MLM13) MLMTAHGBGFNIV
Line 2: (ZUQ08) ZUQUPNZN (ABR14) ABRYCTUGVZXUPB
Line 3: (MVE12) MVERZRLQDBHQ (GKJ11) GKJFHYXODIE
Line 4: (UGM13) UGMNCBXCRLDEY

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Line 5: (HFX19) HFXPCQYZVATXAWIZPVE

Deciphering MLM13 Cipher

With the ring of "NIV," the MLM13 cipher is decoded at the key of "ZNZ" as follows:

00005 - A GOVT CIV BOU CO : AGOVTCIVBOUCO: KEY - ZNZ: RING - NIV

GOVT stands for "GOVERNMENT." CIV stands for "CIVILIAN." BOU stands for "BOUGHT." "CO" stands for "COMPANY." The message implies that the ousted government, as a civilian, bought the shares of the National City Bank.

Deciphering ABR14 Cipher on Line 2

We have already decoded the ZUQ08 cipher, and the recovered text is "XOR ID IWA." It is assumed that the presence of the ZUQ08 cipher in front of the ABR14 cipher implies that the L3Ls for the ABR14 cipher have been replaced ("XORed") with "IWA" before being deciphered. In other words, the ABR14 cipher will be modified from "ABRYCTUGVZXUPB" to "ABRYCTUGVZXIWA" (ABR14a). The new ABR14a cipher is decoded as follows.

00005 - REPAYS PIB L DZ CQ : REPAYSPIBLDZCQ: KEY - ADD: RING - IWA ◇
REPAY SPIBLDZCQ (SPI09)

The SPI09 is then deciphered.

00003 - USO GD WHEN : USOGDWHEN: KEY - VMA: RING - ZCQ

Altogether, the deciphered message of the ABR14a cipher is presented as follows.

ABR14: ABRYCTUGVZXUPB
ABR14a: ABRYCTUGVZXIWA ◇ REPAY US OGD WHEN

"OGD" can be a RING value for another cipher, or it could stand for "OF GOLD." It is noted that if the RING for the ABR14 cipher hadn't been replaced with "IWA," the SPI09 cipher would have had different three letters at the end, and the deciphered result wouldn't be the same as presented here.

Deciphering GKJ11 Cipher on Line 3

The MVE12 cipher in front of GKJ11 is deciphered as "TWO MSG XOR SRH." Nonetheless, there are no two messages to perform the XOR operation on. Instead, "SRH" is used as the RING for the decipherment of the GKJ11 cipher. With the RING of "SRH," the GKJ11 is deciphered as follows.

00004 - A ZH LATE TOLD : AZHLATETOLD: KEY - ZRQ: RING - SRH ◇ AZH LATE
TOLD

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When the L3Ls "DIE" was used as the RING instead of "SRH," the GKJ11 cipher is decoded as follows.

GKJFHYXODIE: AZHLATETXXS -key- KIN -ring- DIE ◇ AZH LATE TXXS

Different RING values produce different results. It is interesting to notice that only three letters at the end of the text are mismatched, not the entire message. Errors in the deciphered text are isolated and not propagated through the remaining part of the text. It can be stated that a one-letter error in the deciphered text can easily be spotted as a typo and would be fixed after the decryption has ended.

Deciphering UGM13 Cipher

With the L3Ls of UGM13 as the RING, the following message was chosen from 17,576 possible outcomes.

00005 - LU QX GUMS SIG CO : LUQXGUMSSIGCO: KEY - JEB: RING - DEY ◇
LUQXGUMS SIG CO

LUQ08: LUQXGUMS

It has a double-encrypted cipher LUQ08 with the RING of "UMS." Two outputs were selected as the plausible deciphered texts.

00003 - ASIAN MX F : ASIANMXF: KEY - MJV: RING - UMS ◇ ASIAN MXF
00003 - US VETS WW : USVETSWW: KEY - ULJ: RING - UMS ◇ US VET SWW

We have two deciphered messages for the UGM13 cipher.

- A. UGM13: UGMNCBXCRLDEY ◇ ASIAN MXF SIG CO
- B. UGM13: UGMNCBXCRLDEY ◇ US VET SWW SIG CO

"ASIAN" implies the people on the oust government side since they were Chinese. If "US VET" were chosen, it would imply that the person named "DEWVRY" is from the YQV16 cipher. "US" could indicate the bank or the United States. "VET" could indicate either a war veteran of the States or a veteran of banking. If "DEWVRY" is a US war veteran, the final deciphered text for the UGM13 cipher would be "US VET SWW SIG CO." Otherwise, "ASIAN MXF SIG CO" can be selected as the recovered text.

Deciphering HFX19 Cipher

It took me a lot of time decoding the HFX19 cipher because using the L3Ls "PVE" as the RING value won't produce any plausible outcomes. This is partly because the length of the HFX19 cipher is relatively long and the readability of deciphered texts is very low. Spotting plausible texts out of 17,576 texts per RING value isn't an easy task. Since the L3Ls of its own cipher were no good, it is time to try the RING value derived from other deciphered texts. Using three letters from deciphered texts as the RING value complicates the decryption process because we don't know which

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one to choose. The working three letters might have been placed inside a cipher that hasn't been decoded yet.

On the right column of the gold bar labeled KO 00803, the inscription line starts as follows:

VIO12: VIOHIKNNGUAB
HFX19: HFXPCQYZVATXAWIZPVE
YQH16: YQHUTABGALLOWLS
XLY20: XLYPISNANIRUSFTFWMIY

Both VIO12 and YQH16 ciphers on the gold bar labeled KO 08124 remain undeciphered because there is no MVE12 cipher next to them.

MVE12: MVERZRLQDBHQ ◇ TWO MSG XOR SRH

The HFX19 cipher was placed between the VIO12 and YQH16 ciphers. This might imply that the RING value of "SRH" applies to the decipherment of the HFX19 cipher. By using "SRH" as the RING, the HFX19 cipher produces the following output:

00009 - US BERGE U IF EK S FG VW TZ : USBERGEUIFEKSFGVWTZ: KEY - HEV:
RING - SRH ◇ US BERGE UIFEKSFGVWTZ

We receive another UIF12 cipher in the decrypted text.

UIF12: UIFEKSFGVWTZ

When the UIF12 cipher was decoded with the RING of "WTZ," we obtained the following outcome:

00005 - GOV YD ZX UFC PC : GOVYDZXUFCPC: KEY - ZQN: RING - WTZ ◇ GOV
YDZXUFCPC

The decrypted text contains another cipher named YDZ09.

YDZ09: YDZXUFCPC

The YDZ09 cipher produces the final text as follows:

00003 - KEY JD SIAA : KEYJDSIAA: KEY - SHZ: RING - CPC ◇ KEY JDS IAA

Altogether, the HFX19 cipher has its final text as follows.

HFX19: HFXPCQYZVATXAWIZPVE ◇ US BERGE GOV KEY JDS IAA

It seems excessive to encrypt a message three times in a row by adding extra text. The result obtained here might be incorrect, and there could be other plausible outcomes. Nonetheless, if the decryption is right, those two KEYS for the government kept by the bank should play a very important role. They might be a KEY-RING pair used for encryption and/or decryption in message exchanges between the government and the bank.

Deciphered Texts on the Bottom Part of Gold Bar with Text Only (Upside Down)

All ciphers on the bottom of the gold bar with dual texts have been listed here with their deciphered texts.

Label	Cipher Message	KEY:RING	Deciphered Message
MLM13	MLMTAHGBGFNIV	ZNZ:NIV	A GOVT CIV BOU CO
ZUQ08	ZUQUPNZN	HQH:FWT	XOR ID IWA
ABR14	ABRYCTUGVZXUPB		(Last 3 letters replaced with IWA)
ABR14a	ABRYCTUGVZXIWA	ADD:IWA	REPAY SPIBLDZCQ ◊ REPAY US OGD WHEN
◊SPI09	SPIBLDZCQ	VMA:ZCQ	US OGD WHEN
MVE12	MVERZRLQDBHQ	GIV:BHQ	TWO MSG XOR SRH
GKJ11	GKJFHYXODIE	ZRQ:SRH	AZH LATE TOLD
UGM13	UGMNCBXCRLDEY	JEB:DEY	LUQXGUMS SIG CO ◊ (a) US VET SWW SIG GO or (b) ASIAN MXF SIG CO
◊LUQ08	LUQXGUMS	ULJ:UMS	(a) US VET SWW
		MJV:UMS	(b) ASIAN MXF
HFX19	HFXPCQYZVATXAWIZPVE	HEV:SRH	US BERGE UIFEKSFGVWTZ ◊ US BERGE GOV KEY JDS IAA
◊UIF12	UIFEKSFGVWTZ	ZQN:WTZ	GOV YDZXUFCPC
◊YDZ09	YDZXUFCPC	SHZ:CPC	KEY JDS IAA

The Right Column of Gold Bar (KO 08124)

The ciphers on the left column have been decrypted earlier. On the right column, there are five ciphers.

MLM13: MLMTAHGBGFNIV ◊ A GOVT CIV BOU CO
 FEW23: FEWGDRHDDEEUMFFTEEMJXZR ◊ OUST GOV P BANK CT GOL EES VOG
 RHZ24: RHZVIYQIYSXVNQXQWIOVWPJO
 HFX19: HFXPCQYZVATXAWIZPVE ◊ US BERGE GOV KEY JDS IAA
 KOW13: KOWVRSRWTMLDH

Three of five ciphers have already been decrypted above.

Deciphering RHZ24 Cipher

With the RING of "PJO", the RHZ24 cipher is decrypted. The deciphered text contains two ciphers and they are NOC13 and PAS06 ciphers.

00009 - PAY NO CNN TTC LAS ES IF PASO LI : PAYNOCNNTTCLASESIFPASOLI:
 KEY - FHK: RING - PJO ◊ PAY NOCNNTTCLASES IF PASOLI

NOC13: NOCNNTTCLASES
 PAS06: PASOLI

The decrypted NOC13 text contains one additional cipher.

00006 - KEY QI AHU NO Q UP : KEYQIAHUNOQUP: KEY - DKU: RING - SES ◊
 KEY QIAHUNOQ UP

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QIA08: QIAHUNOQ

The QIA08 cipher is decrypted.

00003 - IN USS KAN : INUSSKAN: KEY - GGK: RING - NOQ

The decrypted text of NOC13 cipher is "KEY IN USS KAN UP". The PAS06 cipher is decrypted.

00002 - DUE ASK : DUEASK: KEY - YQX: RING - OLI

The final decrypted text of RHZ24 cipher is "PAY KEY IN USS KAN UP IF DUE ASK". The RHZ24 cipher contains a text encrypted three times like HFX19 does. Each RHZ24 and HFX19 ciphers contains a pair of three letter keys.

Deciphering KOW13 Cipher

With the RING of "LDH", the KOW13 cipher is decrypted. It contains another cipher in the decrypted text.

00006 - ACC NON LB RH Q EM : ACCNONLBRHQEM: KEY - NQS: RING - LDH ◊
ACC NO NLBRHQEM

NLB08: NLBRHQEM

The NLB08 cipher is decrypted.

00003 - ASH KEEP I : ASHKEEPI: KEY - OMN: RING - QEM ◊ I KEEP HSA (in reverse)

The final decrypted text is "ACC NO [ASH KEEP I]" or it can be read as "ACC NO [I PEEK HSA]". Remember that the decrypted text of FEW24 cipher also has reversed words such as "GOL EES VOG" at the end and it is read as "GOV SEE LOG".

Deciphered Texts of the Right Column of Gold Bar (KO 08124)

All ciphers on the right column have been listed here with their deciphered texts.

Label	Cipher Message	KEY:RING	Deciphered Message
MLM13	MLMTAHGBGFNIV	ZNZ:NIV	A GOVT CIV BOU CO
FEW23	FEWGDRHDDEEUMFFTEEMJXZR	GPV:XZR	OUST GOV P BANK <u>IZMUBMHU</u> VOG ◊ OUST GOV P BANK CT GOL EES VOG
RHZ24	RHZVIYQIYSXVNQXQWIOVWPJO	FHK:PJO	PAY <u>NOCNNTTCLASES</u> IF <u>PASOLI</u> ◊ PAY KEY IN USS KAN UP IF DUE ASK
◊NOC13	NOCNNTTCLASES	DKU:SES	KEY <u>QIAHUNOQ</u> UP
◊QIA08	QIAHUNOQ	GGK:NOQ	IN USS KAN
◊PAS06	PASOLI	YQX:OLI	DUE ASK
KOW13	KOWVRSRWTMLDH	NQS:LDH	ACC NO <u>NLBRHQEM</u> ◊ ACC NO ASH KEEP

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			I/ACC NO [I PEEK HSA]
ØNLB08	NLBRHQEM	YQX:OLI	ASH KEEP I/I PEEK HSA

The Middle Column of Gold Bar (K00803)

There are five ciphers in the middle column, and they have all been decrypted.

UGM13: UGMNCBXCRLDEY ◇ A GOVT CIV BOU CO
RHZ24: RHZVIYQIYSXVNQXQWIOVWPJO ◇ PAY [KEY IN USS KAN] UP IF DUE ASK
SKC25: SKCDKJCDJCYQSZKTZJPXPWIRN ◇ BAY [ADJ GATA MAST WY] STACKY LOT
MQO24: MQOLCSJTLGAJOKBSSBOMUPCE ◇ USED [JUD BOR] IS GOV ID P KEY SEA
FEW23: FEWGDRHDDEEUMFFTEEMJXZR ◇ OUST GOV P BANK CT [GOL EES VOG]

The Right Column of Gold Bar (K00803)

There is one cipher left to decrypt on the right column of gold bar labeled KO 00803.

VIO12: VIOHIKNNGUAB ◇ (not deciphered)
HFX19: HFXPCQYZVATXAWIZPVE ◇ US BERGE GOV KEY JDS IAA
YQH16: YQHUDTABGALLOWLS ◇ (not deciphered)
XLY20: XLYPISNANIRUSFTFWMIY
KOW13: KOWVRSRWTMLDH ◇ ACC NO ASH KEEP I/ACC NO [I PEEK HSA]
JKG14: JKGFIJPMCWSAEK ◇ ACT DMZ [EFG] SIG CT
ABR14: ABRYCTUGVZXUPB (ABRYCTUGVZXIWA) ◇ REPAY US OGD WHEN
GKJ11: GKJFHYXODIE ◇ AZH LATE TOLD
ZUQ08: ZUQUPNZN ◇ XOR ID IWA
GKJ11: GKJFHYXODIE ◇ AZH LATE TOLD

Deciphering XLY20 Cipher

Since the XLY20 cipher is located below the YQH16 cipher, the ring of "SRH" will be chosen for its decipherment. The following text was selected as the deciphered outcome. It contains two other ciphers.

00010 - QUIZ TW GY X TO A EW MY SX FH : QUIZTWGYXTOAEWMYSXFH: KEY -
DLX: RING - SRH ◇ QUIZ TWGYX TO AEWMYSXFH

TWG05: TWGYX
AEW09: AEWMYSXFH

The TWG05 cipher is decrypted.

00001 - BUYER : BUYER: KEY - BDZ: RING - GYX

The AEW09 cipher is decrypted.

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00003 - BYE ANY OPP : BYEANYOPP: KEY - IKU: RING - XFH

The deciphered text of XLY20 is given as follows.

XLY20: XLYPISNANIRUSFTFWMIY ◊ QUIZ BUYER TO BYE ANY OPP

For the deciphered text of TWG05 cipher, there are a few more alternatives.

00001 - ASHOK : ASHOK: KEY - CIX: RING - GYX ◊ ASH OK (ASH from "ASH
KEEP I" of KOW13 cipher)

00002 - A CLUE : ACLUE: KEY - OLB: RING - GYX

00001 - BYNUM : BYNUM: KEY - TUV: RING - GYX ◊ BY NUM

With all these alternatives, the deciphered text of XLY20 is finalized.

XLY20: XLYPISNANIRUSFTFWMIY ◊ QUIZ BUYER[/ASH OK/A CLUE/BY NUM] TO BYE
ANY OPP

The AEW09 cipher is decrypted. The deciphered text of XLY20 is provided as follows. For the deciphered text of the TWG05 cipher, there are a few more alternatives. With all these alternatives, the deciphered text of XLY20 is finalized. The deciphered text implies that the buyer will be challenged with a quiz to reject any illegitimate opponent.

Full List of the Decrypted CGB Ciphers

Gold Bar (with text only)	KO 08124	KO 00803 (or 00808)
JKGFIJPMCWSAEK (ACT DMZ [EFG] SIG CT)	(LEFT)	(MIDDLE)
SKCDKJCDJCYQSZKTZJPXPWIRN (BAY ADJ GATA MAST WY STACKY LOT)	MQOLCSJTLGAJOKBSSBOMUPCE (USED [JUD BOR] IS GOV ID P KEY SEA)	UGMNCBXCRLDEY (US VET SWW SIG CO)
MQOLCSJTLGAJOKBSSBOMUPCE (USED [JUD BOR] IS GOV ID P KEY SEA)	FEWGDRHDDEEUMFFTEEMJXZR (OUST GOV P BANK CT GOL EES VOG)	RHZVIYQIYSXVNQXQWIOVWPJO (PAY KEY IN USS KAN UP IF DUE ASK)
FEWGDRHDDEEUMFFTEEMJXZR (OUST GOV P BANK CT GOL EES VOG)	YQHUTABGALLOWLS (Not deciphered, see YQV16.)	SKCDKJCDJCYQSZKTZJPXPWIRN (BAY ADJ GATA MAST WY STACKY LOT)
RHZVIYQIYSXVNQXQWIOVWPJO (PAY KEY IN USS KAN UP IF DUE ASK)	VIOHIKNGUAB (Not deciphered, see YQV16.)	MQOLCSJTLGAJOKBSSBOMUPCE (USED [JUD BOR] IS GOV ID P KEY SEA)
MQOLCSJTLGAJOKBSSBOMUPCE (USED [JUD BOR] IS GOV ID P KEY SEA)	MVERZRLQDBHQ (TWO MSG XOR SRH)	FEWGDRHDDEEUMFFTEEMJXZR (OUST GOV P BANK CT GOL EES VOG)
FEWGDRHDDEEUMFFTEEMJXZR (OUST GOV P BANK CT GOL EES VOG)	◇ YQVIOHIKNGUABLS (YQV16) (CONTACT FWT DEWVRY)	(RIGHT)
SKCDKJCDJCYQSZKTZJPXPWIRN (BAY ADJ GATA MAST WY STACKY LOT)	ZUQUPNZN (XOR ID IWA)	VIOHIKNGUAB (Not deciphered, see YQV16.)
RHZVIYQIYSXVNQXQWIOVWPJO (PAY KEY IN USS KAN UP IF DUE ASK)	VIOHIKNGUAB (Not deciphered, see YQV16.)	HFXPCQYZVATXAWIZPVE (US BERGE GOV KEY JDS IAA)
MQOLCSJTLGAJOKBSSBOMUPCE (USED [JUD BOR] IS GOV ID P KEY SEA)	JKGFIJPMCWSAEK (ACT DMZ EFG SIG CT)	YQHUTABGALLOWLS (Not deciphered, see YQV16.)
SKCDKJCDJCYQSZKTZJPXPWIRN (BAY ADJ GATA MAST WY STACKY LOT)	SKCDKJCDJCYQSZKTZJPXPWIRN (BAY ADJ GATA MAST WY STACKY LOT)	XLYPISNANIRUSFTFWMIY (QUIZ BUYER/ASH OK/A CLUE/BY NUM TO BYE ANY OPP)
(UPSIDE DOWN)	(RIGHT)	KOWVRSRWTMLDH (ACC NO ASH KEEP I)
MLMTAHGBGFNIV (A GOVT CIV BOU CO)	MLMTAHGBGFNIV (A GOVT CIV BOU CO)	JKGFIJPMCWSAEK (ACT DMZ EFG SIG CT)
ZUQUPNZN (XOR ID IWA)	FEWGDRHDDEEUMFFTEEMJXZR (OUST GOV P BANK CT GOL EES VOG)	ABRYCTUGVZXUPB (XOR ID IWA) ◇ ABRYCTUGVZXIWA (REPAY US OGD WHEN)
ABRYCTUGVZXUPB (XOR ID IWA) ◇ ABRYCTUGVZXIWA (REPAY US OGD WHEN)	RHZVIYQIYSXVNQXQWIOVWPJO (PAY KEY IN USS KAN UP IF DUE ASK)	GKJFHYXODIE (AZH LATE TOLD)
MVERZRLQDBHQ (TWO MSG XOR SRH)	HFXPCQYZVATXAWIZPVE (US BERGE GOV KEY JDS IAA)	ZUQUPNZN (XOR ID IWA)
GKJFHYXODIE (AZH LATE TOLD)	KOWVRSRWTMLDH (ACC NO ASH KEEP I)	GKJFHYXODIE (AZH LATE TOLD)
UGMNCBXCRLDEY (US VET SWW SIG CO)		
HFXPCQYZVATXAWIZPVE (US BERGE GOV KEY JDS IAA)		

* Unless the original text was known to the public, I believe that the recovered text presented here would be a very close version of it. I must say that the methods I

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discovered might not be the ones actually used in the generation of the CGB ciphers, but I am confident in them.